

# Principles of Quantum Mechanics

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# 1 Introduction

Recall from IB Quantum Mechanics we have the following:

- (i) Wave-particle duality. Waves behaving like particles like light quanta and vice-versa (particles behaving like waves).
- (ii) We have a wavefunction  $\psi(x)$  for a particle which we interpret as a probability density  $|\psi(x)|^2$  (i.e. probability is intrinsic to the theory).
- (iii) Observables become operators on wavefunctions:

- $\hat{x} \rightarrow x$ ;
- $\hat{p} \rightarrow -i\hbar \frac{\partial}{\partial x}$

and the lack of commutation since  $[\hat{x}, \hat{p}] = i\hbar$  leads to the uncertainty relation.

- (iv) Schrödinger's equation with Hamiltonian  $H$  specifies dynamics and eigenvalues of  $H$  are the allowed energy levels of the system.
- (v) All of this explains the hydrogen atom structure and the resulting line spectra.

The aims of this course are the following:

- (i) Reformulate quantum mechanics in a more powerful, flexible and useful form: the Dirac formulation, which goes beyond wavefunctions. The previous description appears as a particular realisation of our generalisation of quantum states and also leads to alternatives such as

$$\tilde{\psi}(p) \quad \text{with } \hat{x} \rightarrow i\hbar \frac{\partial}{\partial p} \quad \text{and } \hat{p} \rightarrow p.$$

- (ii) This more general approach allows much simpler analysis of known problems (for example the Simple Harmonic Oscillator) but also lots of new things such as *spin* of particles and symmetries (translations, rotations) and the related conservation laws.
- (iii) We will also look at identical particles (bosons and fermions).
- (iv) We will develop a framework for quantising more general systems such as EM fields.
- (v) We will also setup perturbation theory which is a practical tool for computation but we will not heavily consider the detailed applications.

## 2 Dirac Formalism

### 2.1 Concepts and Definitions

A quantum system is described at each instant by a *state*  $|\psi\rangle$  which belongs to a complex vector space  $V$ ; any non-zero element of this space  $V$  is a possible state. Thus

$$|\psi\rangle, |\phi\rangle \in V \implies \alpha |\psi\rangle + \beta |\phi\rangle \in V \text{ for any } \alpha, \beta \in \mathbb{C}.$$

Physically this is a superposition principle leading to wave-like behaviour but these states are *not wavefunctions*. There are also *dual* or conjugate states  $\langle\psi|$  which belong to the dual space  $V^\dagger$ . By definition states and duals can be combined or paired to give a complex number. The map

$$\langle\psi|, |\psi\rangle \rightarrow \langle\psi|\psi\rangle$$

defines a map from  $V^\dagger \times V \rightarrow \mathbb{C}$  with

$$\langle \phi | (\alpha_1 | \psi_1 \rangle + \alpha_2 | \psi_2 \rangle) = \alpha_1 \langle \phi | \psi_1 \rangle + \alpha_2 \langle \phi | \psi_2 \rangle$$

and

$$\langle (\beta_1 \langle \phi_1 | + \beta_2 \langle \phi_2 |) | \psi \rangle = \beta_1 \langle \phi_1 | \psi \rangle + \beta_2 \langle \phi_2 | \psi \rangle.$$

The spaces  $V$  and  $V^\dagger$  comes with an inner product which can be described as a one-to-one correspondence between the spaces

$$\begin{aligned} V &\rightarrow V^\dagger \\ | \psi \rangle &\rightarrow \langle \psi | = (| \psi \rangle)^\dagger. \end{aligned}$$

We also have that if we take a linear combination of states to the dual space, the coefficients have to have their complex conjugates taken to from the dual space to  $V$  or visa-versa. Hence the inner product is

$$| \phi \rangle, | \psi \rangle \rightarrow \langle \phi | \psi \rangle = (| \psi \rangle)^\dagger | \phi \rangle$$

and it is assumed to obey

$$\langle \phi | \psi \rangle = \langle \psi | \phi \rangle^*$$

since the inner product is hermitian, and is the inner product defines a positive definite norm on  $V$ .

$$|| | \psi \rangle ||^2 = \langle \psi | \psi \rangle.$$

*Remark.* Knowing  $\langle \phi | \psi \rangle$  for all  $\langle \phi |$  determines  $| \psi \rangle$  and visa-versa.

We can show this by

$$\langle \phi | \psi_1 \rangle = \langle \phi | \psi_2 \rangle \implies \langle \psi | \chi \rangle = 0$$

where  $| \chi \rangle = | \psi_1 \rangle - | \psi_2 \rangle$ . If this holds for all  $\langle \phi |$  we can set  $\langle \phi | = \langle \chi |$  so by positive definiteness we have that the norm of  $| \chi \rangle$  is zero, so  $| \chi \rangle = 0$  too.

We also have that the physically content of any state is unchanged by

$$| \psi \rangle \rightarrow \alpha | \psi \rangle$$

for any  $\alpha \in \mathbb{C}$  non-zero. So we will usually normalise to make the state have norm 1, but we can still change by a phase vector.

*Remark.* Relative phases can be important in expressions where we're adding a linear combination of elements of  $V$ . We also assume the space  $V$  is complete so appropriate sequences and series converge in  $V$ .

A complete inner product space is called a *Hilbert space*.

The same operator can be thought of as acting to the left on dual states, so

$$\langle \phi | \rightarrow \langle \phi | Q$$

where the right hand side is defined by

$$(\langle \phi | Q) | \psi \rangle = \langle \phi | (Q | \psi \rangle) = \langle \phi | Q | \psi \rangle$$

